

ZIA UR REHMAN

Agentic AI Python Backend Engineer

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Career Summary

Results-driven Python Backend Engineer and AI Developer with hands-on experience building agentic AI systems, LLM-powered applications, and scalable REST APIs. Proficient in **LangChain**, **LangGraph**, **FastAPI**, and **Django**, with a strong focus on designing multi-agent workflows, RAG pipelines, and intelligent backend services that solve real-world problems. Passionate about leveraging AI to automate complex processes and deliver impactful, production-ready solutions.

Technical Skills

AI / Agentic AI	LangChain, LangGraph, LangSmith, Agentic Workflows, Multi-Agent Systems, RAG (Retrieval-Augmented Generation), Vector Databases (FAISS, ChromaDB), Prompt Engineering, Tool-Calling Agents, OpenAI API
Backend Frameworks	FastAPI, Django, Django REST Framework (DRF), Python
Databases	PostgreSQL, MySQL, Vector Stores
DevOps / Cloud	Git & Version Control, AWS (Cloud Foundations), Docker (basic)
CS Fundamentals	Data Structures & Algorithms, OOP, Clean Code, Agile/Scrum
Soft Skills	Problem-Solving, Teamwork, Fast Learner, Technical Communication

Work Experience

Software Engineering Intern

March 2025 – April 2025

CodexCue Software Solutions, Pakistan

- Developed and maintained **Python/Django** web applications, optimising backend performance and delivering new features for client-facing projects.
- Integrated **LangChain**-based components to prototype AI-assisted query handling within existing Django services, reducing manual lookup time by over 40%.
- Collaborated in an **Agile** team environment to design, test, and deploy scalable software solutions following industry best practices.
- Wrote clean, testable REST API endpoints using **Django REST Framework**, improving API response consistency across modules.
- Participated in code reviews and contributed to CI/CD pipelines using **Git**.

Projects

MediAssist — AI Medical Chatbot for Patient Health Diagnosis LangChain · FastAPI · PostgreSQL · OpenAI

- Built a **production-ready agentic AI chatbot** that helps patients perform preliminary health assessments by describing symptoms and receiving structured diagnostic guidance, addressing a critical gap in healthcare accessibility.
- Designed a **LangGraph multi-step agent** pipeline: symptom intake → context retrieval (RAG over medical knowledge base) → diagnosis reasoning → specialist referral recommendation, all orchestrated as a stateful graph.
- Implemented **RAG** using **FAISS** vector store over curated medical corpora (symptoms, disease profiles, drug interactions), enabling precise context-aware responses with cited sources.
- Exposed the agent via a **FastAPI** async REST API with JWT authentication, rate limiting, and structured JSON responses; designed for integration with web and mobile front-ends.
- Added **LangSmith** tracing for observability, enabling prompt debugging and latency optimisation across agent nodes.
- Stored patient session history in **PostgreSQL** with full conversation memory using LangChain's

ConversationBufferWindowMemory.

- *Real-world impact:* Reduces unnecessary clinic visits by triaging non-critical symptoms and directing patients to the right specialist, especially valuable in under-resourced regions.

MasjidConnect — SaaS Community Web Platform

Python · Django · PostgreSQL

- Designed and built a multi-tenant **SaaS web application** enabling mosque administrations to manage prayer schedules, community events, and announcements through a dynamic admin dashboard.
- Integrated **auto-sync prayer time APIs** and online Sadqah donation workflows, serving real users across multiple mosque communities.
- Engineered a **LangChain-powered FAQ assistant** as an add-on module, allowing community members to query Islamic Q&A content via natural language over a custom knowledge base.
- Designed **PostgreSQL** schemas for multi-mosque tenancy with role-based access control for admins and regular users.

SialBooks — Full-Stack E-Commerce Bookstore

Python · Django · MySQL

- Developed a fully functional online bookstore with user authentication, email verification, order management, and an admin panel.
- Built a **LangChain tool-calling agent** for book recommendations, allowing users to describe preferences in natural language and receive personalised suggestions from the catalogue.
- Implemented **MySQL** relational models for inventory, orders, and user profiles, with Django ORM optimisations to reduce query count.

Education

BS Computer Science — University of Sargodha

2023 – 2027 (Expected)

GPA: **3.67 / 4.00** Major: Computer Science

FSc. Pre-Medical (Intermediate) — ASPIRE Group of Colleges, Sargodha

Oct 2020 – Jun 2022

Grade: **81%**

Certifications

- **Oracle Certified Generative AI Developer** — Oracle (Sep–Oct 2025)
- **AWS Introduction to Cloud 101** — Amazon Web Services (Jun–Jul 2024)

Extracurricular Activities & Achievements

- **Member**, Techive Computing Society, University of Sargodha (2024–2025)
- **Runner-Up**, Cricket — Intra-Departmental Sports Gala, UOS (2026)
- **Winner**, Debate Competition, Unique Modern School, Sargodha (2019)

References

Available on request.